The Geography of Video Game Genres

Global Patterns of Genre Evolution and Innovation, 1952–2025

Abstract (Provisional)

This study examines the evolution of video-game genre categories through three perspectives: cross-country reach, country–genre specialization, and the emergence of multi-genre combinations. The IGDB-derived dataset covers 75,971 games released between 1952 and 2025, 118 developer-country identifiers, and 23 genre labels. For diffusion, a country–genre appearance is defined as an annual fractional weighted count of at least three games. Cumulative reach curves are constructed descriptively; no parametric logistic or Bass curve is fitted. Sustained decline is flagged when annual counts remain below 25% of peak for five consecutive years. Under the adoption threshold, Indie appears in 69 developer-country portfolios and Adventure in 67; Simulator has the earliest first threshold year among these leading genres (1972). Country specialization is measured using five-year Revealed Comparative Advantage (RCA). Next-period specialization rates rise from 7.4% in the lowest relatedness quartile to 28.6% in the highest. However, only 16 countries receive a substantive trajectory label; 102 have insufficient period coverage. In link prediction, global common-neighbor ranking identifies five of six previously unseen pairs observed in 2016–2025 but does not outperform a random baseline convincingly ($p = 0.5594$). A preliminary country-specific evaluation achieves 0.632 precision across 470 top-ranked slots ($p = 0.000999$). The findings describe database associations, not causal diffusion or consumer adoption; RCA represents relative metadata specialization, not national comparative advantage.

Keywords: video-game genres; innovation diffusion; economic complexity; genre relatedness; network science; link prediction

1. Introduction

Video games are a globally distributed cultural and commercial form whose products are classified through categories such as action, adventure, role-playing, simulation, and strategy. These labels make a heterogeneous population of games searchable and comparable, but they do not describe mutually exclusive types. A single title can combine several genres, and the meaning or prominence of a category can change over time. Genre is therefore both a descriptive attribute of individual products and a classification system whose structure can be studied historically. Research on game genre has emphasized the limitations of fixed taxonomies and the value of treating games as multidimensional cultural objects (Apperley, 2006; Faisal & Peltoniemi, 2018). Data-driven analyses of tags and labels further demonstrate that category structures can be reconstructed from large product databases, while remaining dependent on the classification practices of the underlying source (Li, 2020; Starosta et al., 2024; Grelier et al., 2025).

This study approaches genre categories as evolving, overlapping labels and examines their recorded appearance across time and country associations. Its starting point is the literature on innovation diffusion. Rogers (2003) conceptualizes diffusion as a process unfolding over time within a social system, drawing attention to the innovation, communication channels, time, and the population among which an

innovation becomes known. In the present empirical setting, the available data permit the observation of temporal appearances and their cumulative reach, but not the direct observation of communication between developers, firms, or audiences. Accordingly, the paper uses threshold-based first appearance and cumulative reach as descriptive indicators. These indicators can reveal whether a category becomes represented in a wider set of country-associated records, but they do not establish where an innovation originated or how it was transmitted. Temporal precedence is not equivalent to influence: a country-associated record appearing earlier than another may reflect differences in metadata coverage, company attribution, or release documentation as well as a genuine difference in timing.

A second perspective comes from economic complexity and the product-space literature. In this research tradition, the structure of an economy's existing portfolio is associated with the set of products into which it subsequently diversifies. Revealed Comparative Advantage (RCA) is used to identify products that occupy a larger share of a country's export basket than of the global basket (Balassa, 1965). Relatedness measures then represent the proximity between products, with countries tending to enter activities connected to their existing productive structure (Hidalgo et al., 2007; Hausmann & Klinger, 2007). This study adapts that portfolio logic to country-associated game–genre records. The adaptation is deliberately limited: the data do not measure exports, value added, employment, or national capabilities. RCA is therefore interpreted as relative genre representation within the observed metadata, not as a full measure of economic comparative advantage. The key question is whether country–genre records that are more closely related to a country's existing specializations are more likely to cross the study's specialization threshold in a subsequent period.

A third perspective concerns recombination and networks. Innovation may involve the reconfiguration of established components rather than the introduction of entirely new components (Henderson & Clark, 1990). Work on technology brokering similarly emphasizes the recombination of knowledge across domains (Hargadon & Sutton, 1997), while network research provides methods for describing relations among categories and ranking links that have not yet appeared (Adamic & Adar, 2003; Liben-Nowell & Kleinberg, 2007). In this study, genre labels are represented as nodes and their co-occurrence within games as edges. The resulting network supports analysis of genre proximity and a temporal link-prediction task: whether network structure observed up to a cutoff year can rank genre pairs that first appear in a held-out period. Because the global genre vocabulary contains only 23 labels, most possible pairs have already appeared by 2015. The paper therefore also considers triples and country-specific pairs, where the candidate spaces are less saturated.

The empirical study uses an IGDB-derived relational dataset covering 1952–2025. The retained analytical sample contains 75,971 games, 118 country identifiers, and 23 genre labels. Each game can be linked to multiple genres and, through developer-company relationships, to more than one country. Country associations are deduplicated at the game–country level and fractionally weighted when a game is associated with multiple distinct developer countries. Genre labels remain overlapping: a multi-genre title contributes to each of its recorded genres and to the corresponding pair and triple combinations. This construction allows the analysis to move between annual country–genre counts, five-year country–genre portfolios, and game-level genre co-occurrence networks.

Three research questions guide the analysis. RQ1 asks how the country-associated representation of genre categories changes over time: when do genres first cross a consistent annual threshold in different countries, how widely do they become represented, and what differences in reach and persistence are observable? RQ2 asks which country–genre combinations exhibit relative specialization within the dataset

and whether a genre's relatedness to a country's existing specializations is associated with next-period entry. RQ3 asks whether observed genre co-occurrence structure can rank previously unseen pairs, triples, and country-specific pairs that appear in a later period. These questions connect diffusion analysis, economic-complexity-inspired portfolio measures, and network prediction within one longitudinal design.

The reported findings are mixed and should be interpreted in relation to the measurement design. Under the annual threshold of at least three fractional weighted game records, Indie reaches 69 country-associated portfolios and Adventure reaches 67; Simulator has the earliest first threshold year among the leading categories, in 1972. The relatedness analysis shows a monotonic increase in next-period specialization rates, from 7.4% in the lowest relatedness quartile to 28.6% in the highest. At the same time, only 16 of the 118 represented countries receive a substantive trajectory classification under the minimum-coverage rule, while 102 lack sufficient period coverage. In the link-prediction experiment, global pair ranking does not significantly outperform the random baseline, whereas a country-specific version yields a stronger preliminary result that requires additional robustness checks. The analysis therefore emphasizes both positive associations and null or inconclusive predictive evidence.

The study makes three contributions. First, it provides a longitudinal description of genre reach and temporal representation across country-associated records. Second, it transfers the logic of revealed specialization and relatedness to a multi-label cultural-product dataset, while making clear the difference between metadata representation and economic production. Third, it evaluates network-based prediction of genre recombination under a temporal holdout, distinguishing the saturated global pair problem from country-conditional novelty. The paper does not test the geographic-distance, shared-language, publisher-network, platform, or market-size mechanisms proposed in the initial research proposal because the available analytical variables do not provide validated measures of those mechanisms. Work on global game production networks and platformized cultural production motivates these questions, but also underscores the importance of identifying the actual firms, institutions, and infrastructures through which production and circulation occur (Johns, 2006; de Vaan et al., 2013; Kerr, 2017; Poell et al., 2021).

The remainder of the paper is organized as follows. Section 2 reviews the literature on genre classification, innovation diffusion, economic complexity, network prediction, and the geography of cultural production. Section 3 describes the IGDB-derived source data, the construction of developer-country and genre associations, the sample-selection process, and the resulting analytical dataset. Section 4 presents the computational methods and operational definitions. Section 5 reports the empirical results for genre reach, country specialization and relatedness, and genre-combination prediction. Section 6 discusses the findings in relation to the theoretical framework. Section 7 sets out the limitations and priorities for future research, and Section 8 concludes.

2. Literature Review and Theoretical Background

This study brings together five literatures that are often treated separately: research on genre and classification, work on innovation diffusion and cultural evolution, economic complexity and relatedness, network-based link prediction, and the geography and platformization of cultural production. Their integration provides a framework for examining genre labels not as fixed attributes of games, but as observable categories that change in prevalence, combine with other categories, and become differentially associated with national settings over time. The empirical analysis is deliberately narrower than a full

account of cross-border knowledge transfer: it observes metadata associations and their temporal structure, rather than the organizations or audiences through which ideas may travel.

2.1. Genre as a changing and multidimensional classification system

Classification is not a neutral preliminary step in the analysis of cultural products. Categories make heterogeneous objects comparable, but they also impose boundaries that can conceal overlap, hybridity, and change. Apperley (2006) argued that market-based genre labels flatten differences in play and interaction, and Clarke et al. (2017) analyzed why genre schemes for games repeatedly fail as classification systems. In the sociology of categories, labels are understood as shared concepts that guide how producers and audiences interpret objects (Hannan et al., 2019). A database genre therefore records a classification practice rather than an exhaustive description of a game. Some labels are not descriptions of play at all: 'Indie' conflates several senses of independence and is closer to a market or organizational designation than to a play form (Garda & Grabarczyk, 2016).

A recent data-driven literature responds by reconstructing categories from large databases. Faisal and Peltoniemi (2018) rebuilt genre patterns from product databases, found that classifications differ between sources, and showed that many titles carry more than one genre. Li (2020) treated genre as a latent structure recoverable from Steam tags, and Li and Zhang (2020) analyzed the co-occurrence network of those tags through centrality and community detection. Starosta et al. (2024) documented how inconsistently games are classified across psychological research. Longitudinal studies add the temporal dimension. Grelier et al. (2025) used Steam tag histories to separate short-lived fads, contemporary fashions, and stable classics, and estimated that a trend surge lasts about four years on average; Grelier et al. (2026) applied topological data analysis to the structural evolution of Steam genres, using Simulation games as a case study. In the most explicitly evolutionary account, Valverde et al. (2025) reconstructed the trajectories of more than 7,000 arcade machines over four decades and described genres as niches that diversify, stabilize, and, in some cases, collapse.

These studies support the use of longitudinal metadata but warn that observed patterns are shaped by the taxonomy and the platform from which they are drawn. Gap. Existing work analyzes a single platform, a single hardware niche, or a single genre family. It rarely links the prevalence of genres, their combinations, and their association with countries within one multi-country, multi-decade dataset.

2.2. Innovation diffusion, recombination, and temporal precedence

Diffusion theory defines diffusion as the communication of an innovation through channels, over time, among the members of a social system (Rogers, 2003). More recent reviews continue to stress that adoption depends on the attributes of the innovation, on communication networks, and on context (Dearing & Cox, 2018). Parametric S-curves, of which the Bass model is the best known, have been reviewed extensively (Meade & Islam, 2006), but they presuppose well-defined adopters and a saturation ceiling. Where these are uncertain, cumulative reach under a transparent adoption rule is a more cautious descriptor. Cross-country evidence shows that technologies reach countries at very different times (Comin & Hobijn, 2010), and that lags in adoption should be distinguished from differences in the intensity of use (Comin & Mestieri, 2018). Network theory suggests a mechanism: weak ties carry information between groups (Granovetter, 1973), yet behaviors that require social reinforcement spread better through clustered ties than through bridges alone (Centola, 2018).

Many innovations in categorized markets are recombinations. Studies of science and patents locate novelty in unusual combinations of conventional components (Uzzi et al., 2013; Youn et al., 2015) and associate novel combinations with higher impact but also with greater risk (Veugelers & Wang, 2019). Evidence from cultural markets is consistent: popularity in music is highest for products that are moderately differentiated from their peers (Askin & Mauskapf, 2017), and recombination in hip hop has been linked to market impact (Mueller, 2022). Klement and Strambach (2019) modeled urban music scenes as symbolic knowledge bases and found that new genres arise mainly from local knowledge at the center of scenes, but rarely without extra-local contributions.

For the present dataset, temporal precedence must be separated from influence. If country-associated records for a genre cross an operational threshold before those of another country, the ordering is evidence of earlier metadata appearance under the chosen rule. It is not proof that one country influenced the other, that developers copied one another, or that an audience adopted a new form. The empirical strategy therefore uses 'first appearance,' 'threshold-based entry,' and 'follower-like timing' as bounded descriptive terms. Gap. Diffusion research has supplied concepts for temporal spread, and recombination research has supplied concepts for novelty, but few studies construct comparable country-by-genre timelines and ask whether timing patterns are systematic across categories.

2.3. Economic complexity, revealed specialization, and relatedness

Balassa's (1965) revealed comparative advantage (RCA) compares the share of an activity in a country's output with its share in the world. It is widely used as a relative specialization measure, although it is sensitive to small denominators and skewed distributions (Laursen, 2015), and it is not a measure of productivity, scale, or quality. The product-space approach connects activities by the extent to which the same places specialize in both, and shows that countries tend to move into activities close to those they already perform (Hidalgo et al., 2007; Hausmann & Klinger, 2007). This pattern has since been consolidated as the principle of relatedness (Hidalgo et al., 2018; Boschma, 2017), extended to regions, cities, and firms (Neffke et al., 2011), and reviewed as a new paradigm of economic complexity (Balland et al., 2022; Hidalgo, 2021). Recent work refines the principle in several directions. Relatedness itself changes over time with co-location and complexity (Juhász et al., 2021); countries are more likely to enter activities that their neighbors already export (Bahar et al., 2014); and entry into unrelated activities is less frequent but not absent (Pinheiro et al., 2022). The taxonomy used to define activities shapes the resulting structure (Zaccaria et al., 2014), complexity indices require careful interpretation (Mealy et al., 2019), and the framework now informs policy (Balland et al., 2019; Hidalgo, 2023).

The logic has begun to move to cultural and creative production. Klement and Strambach (2019) test whether new music genres are related or unrelated to a scene's existing knowledge base, and the World Intellectual Property Organization (WIPO, 2024) examines relatedness and local know-how in the development of video game hubs.

The present study adapts this logic to country-associated game-genre records, with one conceptual translation that must remain explicit. The dataset does not measure exports, value added, employment, capabilities, or trade flows. RCA therefore indicates relative representation within the database, not national comparative advantage in the economic sense. Gap. Relatedness has mostly been tested on trade, patent, and industry data with stable classifications. It has seldom been tested on a multi-label cultural taxonomy, in which the same product can belong to several categories at once and in which metadata coverage varies by country.

2.4. Co-occurrence networks and link prediction

In a genre co-occurrence network, genres are nodes and an edge indicates that two labels appear on the same game. Weighting by pointwise mutual information (PMI) or Jaccard similarity distinguishes frequent co-occurrence from co-occurrence that is strong relative to marginal frequency, and community detection then groups labels with similar neighborhoods (Li & Zhang, 2020). Link prediction asks whether a network observed up to a given date can rank the links that form later (Liben-Nowell & Kleinberg, 2007). Common Neighbors, Jaccard, Adamic–Adar, and preferential attachment are standard similarity heuristics (Adamic & Adar, 2003; Newman, 2001), summarized in surveys (Lü & Zhou, 2011; Kumar et al., 2020). They are rankings, not calibrated probabilities, and evaluation choices matter: the imbalance between positive and negative candidates and the baseline against which precision is judged can change conclusions (Yang et al., 2015). Learned approaches such as graph neural networks outperform heuristics in many settings (Zhang & Chen, 2018), and in economic complexity machine learning has been used to forecast the products that countries will add (Albora et al., 2023).

Two features of the genre setting differ from the usual application. First, a small vocabulary saturates: as co-occurrence links accumulate, few unseen global pairs remain. Second, games carry more than two labels, which motivates higher-order approaches that predict the closure of triples and larger sets (Benson et al., 2018; Battiston et al., 2020). Gap. How far simple topological similarity can predict new category combinations under temporal validation, for global pairs, for triples, and within country-specific portfolios, has not been tested.

2.5. Global cultural production, geography, and platformization

Research on the video-game industry shows that cultural products are created and circulated through geographically uneven production systems. Johns (2006) examined value capture, power, and embeddedness in game production networks; De Vaan et al. (2013) studied clustering and firm performance in the global industry; Kerr (2017) analyzed its changing geography, organization, and circulation; and Wolf (2015) collected national and regional histories. Platform research emphasizes that digital platforms reshape the infrastructure, governance, and visibility of cultural production (Nieborg & Poell, 2018; Poell et al., 2021). Cultural-flow scholarship adds that circulation is conditioned by cultural proximity rather than distance alone (Straubhaar, 1991; Iwabuchi, 2002). These studies show why country and region are important analytical dimensions, but they also show that the relevant units may be firms, workers, publishers, and platforms rather than the country labels attached to titles.

A further body of work concerns the data themselves. Digital-trace and crowdsourced metadata inherit selection, coverage, and measurement biases (Olteanu et al., 2019), and descriptive claims should be bounded by what was actually recorded (Salganik, 2018). The IGDB schema does not contain validated measures of shared language, geographic distance, firm or publisher relationships, platform histories, or consumer exposure. The study therefore retains geography as a dimension of metadata association and does not test whether proximity of any kind causes diffusion. Gap. Country-level genre dynamics in games have mostly been studied through case studies and clusters, not through a common longitudinal measure of country–genre representation.

2.6. Synthesis and research questions

Taken together, the literatures suggest that genre categories are historically contingent, multi-label classifications; that innovation spreads through social and organizational channels and often takes the form of recombination; that portfolio diversification may be conditioned by relatedness; and that network topology can provide signals about future co-occurrence. The present design integrates these perspectives through three empirical questions:

RQ1 — Diffusion and reach: How do genre-associated records first appear across countries over time, how widely do genres become represented under a consistent annual threshold, and what differences in temporal reach and persistence are observable across genres?

RQ2 — Specialization and relatedness: Which country–genre combinations display relative specialization within the IGDB sample, and is a country more likely to enter a new genre in the next five-year period when that genre is more closely related to its existing specialties?

RQ3 — Recombination and prediction: To what extent can prior co-occurrence structure rank genre pairs, triples, and country-specific pairs that appear for the first time in a held-out period?

These questions are deliberately framed around observed metadata patterns. They do not assert that country is the place of development, that the first country-associated record is the origin of an innovation, or that later appearances result from direct imitation. The contribution lies in bringing diffusion-style timelines, economic-complexity-inspired specialization measures, and temporally validated network prediction into one reproducible analysis of a large longitudinal cultural-product dataset.

3. Data

3.1. Data source and relational structure

The analysis uses an IGDB-derived relational SQLite database assembled from game records and their associated company, country, genre, keyword, and platform relations. The original source database contains the tables games, companies, countries, game_companies, genres, game_genres, keywords, game_keywords, platforms, and game_platforms. The analytical dataset is not a direct copy of those source tables: it is constructed by joining the relevant entities and bridge tables, applying the inclusion rules described below, and creating a game–country bridge from developer-company relationships.

The raw schema contains a game_companies bridge with game_id, company_id, is_developer, and is_publisher fields. The companies table contains country_id, which links to the countries lookup. For this study, the country association is defined using developer companies only (is_developer = 1); publisher-only associations are not used to assign country. Thus, the source relationship is games → game_companies → companies → countries, filtered to developer-company links. The derived analytical bridge game_countries(game_id, country_id) is created from this relation; it is not a raw table in the original database. Genre associations are derived from game_genres joined to genres.

Table 3.1. Principal source tables and their analytical roles

Table	Principal fields	Role in this study
games	id; name; first_release_date; release_year; ratings fields	Game-level record and release-year anchor.

game_companies	game_id; company_id; is_developer; is_publisher	Identifies developer and publisher company links; developer flag defines country attribution.
companies	id; name; country_id	Company lookup and country link for developer-company associations.
countries	id; name	Country lookup.
game_genres	game_id; genre_id	Game–genre bridge.
genres	id; name	Canonical set of 23 genre labels in the retained sample.
game_platforms / platforms	game_id; platform_id / id; name	Platform associations available in source data but not used in the current temporal models.
game_keywords / keywords	game_id; keyword_id / id; name	Keyword associations available in source data but not used in the current 23-genre analysis.
Derived game_countries	game_id; country_id	Deduplicated game–developer-country bridge derived from developer company links.

Note. The original database contains approximately 269.6 thousand game records, 52.1 thousand companies, 249 country entries, 255.2 thousand game–company links, 487.1 thousand game–genre links, 585.7 thousand game–keyword links, 458.9 thousand game–platform links, 23 genres, 6.3 thousand keywords, and 217 platforms. These source-table row counts are rounded inventory figures; the exact sample-flow counts below are distinct-game counts.

3.2. Country-association construction and duplicate handling

Country is operationalized as the country recorded for a company linked to a game as a developer. The extraction filters game_companies to rows with is_developer = 1, joins company_id to companies.id, and joins companies.country_id to countries.id. A company link without a valid country identifier, or one that does not resolve to a country record, does not create a valid country association. Publisher-only company links are excluded from this country derivation. The resulting country variable should therefore be described as the country of the associated developer company in the database, not as the nationality of individual developers or definitive proof of the physical location where all development work occurred.

The join can generate repeated game–country links when several developer companies associated with the same game are registered in the same country. These repetitions are removed by retaining distinct (game_id, country_id) pairs. Consequently, multiple developer companies in one country contribute one country association for that game, rather than multiple credits. If a game is associated with developer companies in k distinct countries after this deduplication, each country receives a fractional weight of 1/k. The weights for a game therefore sum to one across its distinct developer-country associations. A game associated with two distinct developer countries contributes 0.5 to each; a game associated with one country contributes 1.0.

This rule separates company multiplicity from country multiplicity. Multiple companies in the same country do not inflate the country count, while genuine multi-country developer associations are retained but fractionally weighted. The manuscript's country-based estimates use these fractional weights where specified; raw distinct-game counts are also retained for descriptive reporting. The publisher flag is available in the source schema but is not combined with the developer flag in the main country definition.

Any alternative rule—publisher-only or developer-or-publisher—would define a different analytical population and should be treated as a sensitivity analysis rather than silently mixed with the primary sample.

3.3. Sample selection and retained analytical corpus

The sample-selection flow is sequential. Starting from 269,681 distinct records in the source games table, four records are removed because they lack a valid integer release year within the study window of 1952–2025, leaving 269,677 games. The developer-country join then retains 80,426 games with at least one valid developer-country association; 189,251 are removed at this stage. Finally, the genre join retains 75,971 games with at least one valid genre association, removing a further 4,455 games. The final sample therefore consists of 75,971 unique games.

Table 3.2. Overview of the retained research dataset

Selection stage	Games retained	Excluded at this stage
Raw games in source games table	269,681	—
Valid integer release year, 1952–2025	269,677	4
At least one valid developer-country association	80,426	189,251
At least one valid genre association (final sample)	75,971	4,455

Note. Exclusions are sequential: each exclusion count is calculated among games surviving the previous stage. These values should not be summed with overlapping missingness counts from a separate missing-data audit.

The retained sample contains 118 distinct developer-country identifiers and 23 genre labels. After deduplicating game–country and game–genre pairs, the relational join produces 158,939 distinct game–country–genre rows. The game–genre associations yield 130,657 observed unordered pair events and 73,984 observed unordered triple events. With 23 genre labels, the theoretical candidate spaces contain 253 possible unordered pairs and 1,771 possible unordered triples.

Table 3.2. Overview of the retained research dataset

Measure	Value	Interpretation
Study window	1952–2025	74 calendar years inclusive.
Eligible unique games	75,971	Games with year, country association, and at least one genre.
Countries represented	118	Distinct country labels represented in the analytical records.
Genres represented	23	Distinct genre labels.
Game–country–genre rows	158,939	Joined long-format association records.
Game–genre pair events	130,657	Observed game-level two-genre combinations.
Game–genre triple events	73,984	Observed game-level three-genre combinations.

Derived game–country links	77,157	Distinct game–developer-country bridge records.
Derived game–genre links	156,277	Distinct game–genre bridge records.
Possible unordered pairs	253	23 choose 2.
Possible unordered triples	1,771	23 choose 3.

Note. Pair and triple events are not independent observations. A game tagged with four genres contributes six unordered pairs and four unordered triples.

3.4. Multi-country weighting and multi-genre classification

Let C_i denote the set of distinct developer countries linked to game i after deduplication, and let k_i = |C_i|. The fractional country weight assigned to country c for game i is:

$$w_{ic} = 1 / k_i, \quad \text{for } c \in C_i$$

The country weights for each game sum to one. A weighted country–genre–year count is obtained by summing the relevant game weights over games associated with the genre and country in that year. This prevents a title linked to several developer countries from contributing more than one total country credit across those countries.

Genre associations are treated as multi-label rather than mutually exclusive. Distinct (game_id, genre_id) pairs are retained, but a game's contribution is not divided by the number of genres assigned to it. A title with several genre labels therefore contributes to each label. This is necessary for studying combinations, although it means that genre-specific totals may sum to more than the number of unique games. For the global genre co-occurrence network, each eligible game contributes once to each distinct genre and to each unique pair or triple implied by its label set.

3.5. Platform, keyword, and company information not used in the main models

The original database also contains platform and keyword associations. These fields are retained as potentially valuable extensions but are not used as explanatory variables in the present analyses. In particular, the game_platforms bridge records platform associations but does not, in the supplied schema, distinguish reliably between a game's original platform at first release and platforms added through later ports, remasters, re-releases, or backward compatibility. Assigning every listed platform to the game's original release year could therefore create historically misleading platform exposure. Platform analysis requires a platform-specific release chronology or a defensible rule for identifying first release versus subsequent availability.

The source also contains publisher flags and company identifiers. The present study uses developer-company country to define the primary country association, but it does not construct inter-firm collaboration networks, ownership networks, or publisher-mediated diffusion pathways. Such analyses would require more detailed and validated relational data, including the temporal nature of company relationships. Similarly, although keywords are available, the current paper focuses on the 23 canonical IGDB genre labels; keywords are not treated as interchangeable with genres.

3.6. Coverage, selection, and measurement considerations

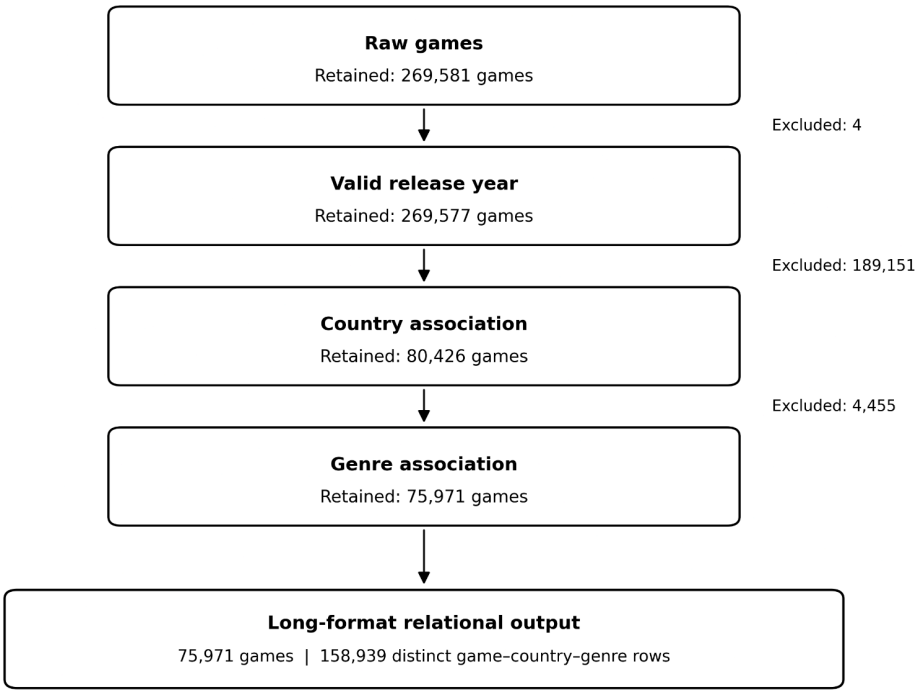
The 75,971-game corpus is a metadata-defined subset of the 269,681 raw game records, not a census of all games developed or played worldwide. The principal filters exclude games without a valid release year, without a valid developer-company country association, or without a valid genre association. The country filter is particularly consequential: games developed by individuals or companies without a recorded country, and games whose developer-company relationship is absent or incomplete, cannot contribute to the country-level analyses. The retained records may therefore overrepresent titles with better documented organizational metadata. Missing country information should not be interpreted as absence of development activity in a country.

Historical coverage may also vary across decades. Earlier games may be less completely documented than more recent releases, while the growth of digital cataloguing may increase the likelihood that recent games have detailed genre and company associations. Consequently, temporal increases in recorded genre counts can reflect both changes in the underlying population of games and changes in database coverage. Thresholds, minimum counts, and relative measures are reported to make the operational definitions explicit, but they do not remove selection bias.

Finally, the country variable measures the country assigned to a linked developer company in the source metadata. It does not directly identify the nationality of creative personnel, the location of each development studio, the place where a game was designed, the market in which it was sold, or the country in which it was played. The dataset contains no comprehensive game-level country-of-play, sales, active-user, download, or playtime series. Thus, terms such as “consumer adoption,” “market penetration,” and “national popularity” are not warranted by these data. The diffusion analysis is more precisely interpreted as the spread of repeated country-associated metadata appearances under an explicit threshold.

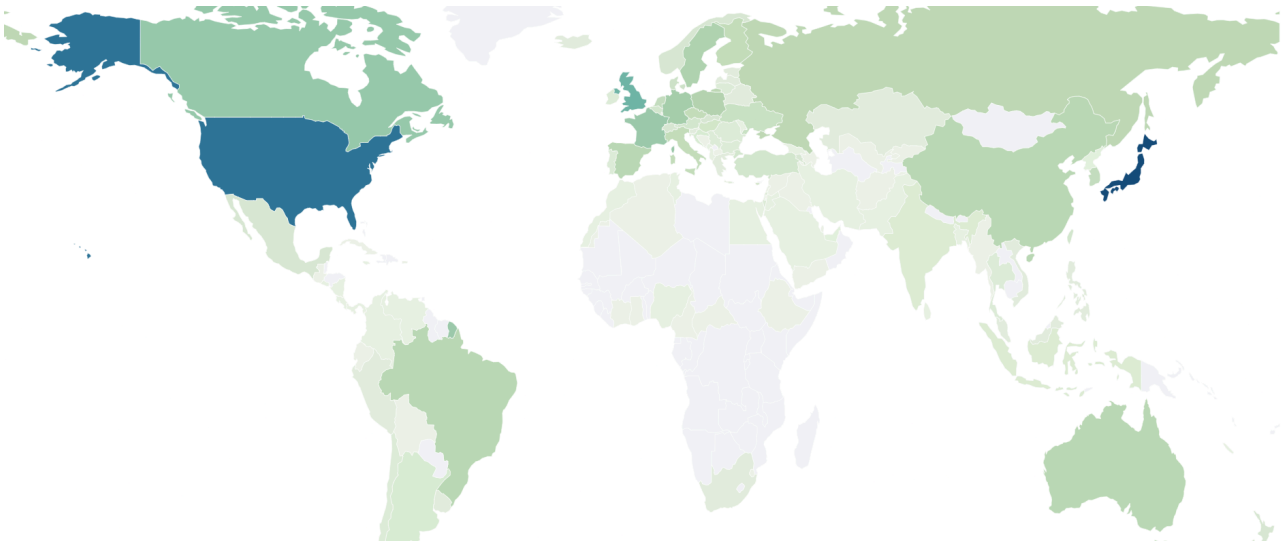
These limitations define the scope of inference. The study can compare the structure and timing of recorded genre associations, the relative composition of developer-country portfolios, and the subsequent appearance of genre combinations. It cannot, without additional data, identify the causal channels through which a genre design travels or establish when audiences in particular countries begin to use it.

Figure 3.1. Sample selection and relational data construction



Country rule: developer; valid release-year window: 1952–2025. Exclusions are sequential.

Figure: Spatial Concentration of Cumulative Video Game Releases Across International Developer Ecosystems, 1952–2025 ($N = 75,971$).



Intensified color gradations reflect higher cumulative software output volumes, illustrating structural asymmetries between dominant global production centers and peripheral markets.

4. Methods

The study uses a sequence of descriptive, comparative, and predictive analyses. Each analytical pillar uses a different representation of the same underlying metadata: country–genre–year counts for diffusion; country–genre–period shares for specialization; country-level specialization sets for relatedness; and genre co-occurrence graphs for combination prediction. The methods below distinguish the procedures implemented in the notebooks from extensions that would require additional modelling or data.

4.1. Construction of the weighted analytical panel

Let i index games, c countries, g genres, and t years or five-year periods. Let C_i denote the set of country associations recorded for game i, and let $k_i = |C_i|$. The fractional country weight is:

Equation (1). Fractional country weight

$$w_{ic} = 1 / k_i \quad \text{for } c \in C_i$$

Each game contributes one total country credit across its linked countries. A game linked to two countries contributes 0.5 to each; a game linked to four contributes 0.25 to each.

The annual weighted count for a country–genre–year is the sum of these weights over all games i associated with country c and genre g in year t. The data are also aggregated into five-year periods beginning in 1952. Duplicate game–country and game–genre associations are removed before joining, and a distinct game-country-period table is used for country denominators so that a game is not multiplied by its number of genre labels.

Equation (2). Annual weighted country–genre count

$$X(c,g,t) = \Sigma_i \, 1(year_i = t) \cdot 1(g \in G_i) \cdot 1(c \in C_i) \cdot w_{ic}$$

The indicator functions select games released in year t and associated with genre g and country c. The corresponding raw measure is the number of distinct game identifiers in the same cell.

4.2. Genre diffusion and country-associated first appearance

The diffusion analysis begins with annual counts by genre and country. A country–genre pair is classified as having reached the operational adoption threshold in the first year for which the annual weighted count is at least three. The threshold is tested at two and five weighted game credits as a sensitivity analysis. The first-adoption year is therefore a threshold-crossing date, not necessarily the date of the first game carrying that genre label.

For each genre, the cumulative reach curve counts the number of countries whose first-adoption year has occurred by year t. Summary measures include the first threshold-crossing year, the total number of countries reaching the threshold, the number reaching it within ten years of the first country, and the elapsed time from the first to the fifth country. These measures describe reach and speed in the recorded metadata. The notebooks also compute annual peaks and a rule-based decline measure: after the peak year, a genre is flagged when its annual weighted count remains strictly below 25% of its peak for five consecutive calendar years. The first year of that five-year run is recorded as the decline year. This rule detects sustained relative contraction; it does not imply extinction or the disappearance of a genre from practice.

A transparent typology further distinguishes long decline (the five-year low-activity rule), local/limited spread (fewer than five adopting countries), fast global spread (at least ten countries crossing the threshold within ten years of the first), slow spread (at least ten countries eventually crossing), and an

intermediate category. These are descriptive labels generated by pre-set rules, not latent classes discovered by a statistical model.

The country lead–lag analysis compares first-adoption dates across genres. A directed lead relationship A → B is considered when A reaches the threshold one to three years before B. The notebook restricts comparisons to country pairs sharing at least five adopted genres and uses a 70% lead-share threshold to identify stronger directional patterns. The result is a temporal precedence network, not an influence network in the causal sense. The cumulative adoption curves in the current notebook are descriptive; a logistic or Bass diffusion curve has not been fitted in the reported output. Likewise, Cox survival models, NetInf, dynamic time warping, and gradient-boosted adoption prediction are not reported as executed analyses.

4.3. Revealed specialization and country–genre portfolios

Country specialization is measured in five-year periods anchored at 1952: 1952–1956, 1957–1961, and subsequent five-year blocks. The final block, 2022–2026, is partial because the data end in 2025; the latest complete period used for the principal summary is 2017–2021. Let $X(c,g,p)$ be the weighted count of genre g in country c in period p, $X(c,\cdot,p)$ the country's total weighted game credits in that period, $X(\cdot,g,p)$ the global weighted count for genre g, and $X(\cdot,\cdot,p)$ the global weighted game total. RCA is defined as:

Equation (3). Revealed comparative advantage (RCA)

$$RCA(c,g,p) = [X(c,g,p) / X(c,\cdot,p)] / [X(\cdot,g,p) / X(\cdot,\cdot,p)]$$

RCA above one means genre g occupies a larger share of country c's recorded genre credits than it does in the full dataset for the same period. Here it measures relative representation in IGDB records, not export competitiveness or national comparative advantage in a full economic sense.

A country–genre–period is designated a specialization only when RCA exceeds one and the weighted game count is at least three. Retaining both conditions reduces the influence of extremely small cells in which one or two records can generate a very large ratio. Raw and weighted counts are reported with RCA values, because a high ratio can occur for a niche genre with a small absolute count.

4.4. Genre co-occurrence, families, and relatedness

The global genre co-occurrence analysis uses unique game–genre associations, so country multiplicity does not duplicate a game in the co-occurrence denominator. Let N be the number of eligible games, nA the number with genre A, nB the number with genre B, and nAB the number carrying both. Pointwise mutual information measures the difference between observed joint frequency and the frequency expected under independence:

Equation (4). Pointwise mutual information

$$PMI(A,B) = \log_2 \{ P(A,B) / [P(A)P(B)] \}$$

Positive PMI indicates co-occurrence above the independence expectation; negative values indicate co-occurrence below it. Normalized PMI scales PMI by the surprisal of the joint event. PMI can be sensitive to rare events, so raw co-occurrence counts and Jaccard similarity are inspected alongside it.

Equation (5). Jaccard similarity for genre game-sets

$$J(A,B) = nAB / (nA + nB - nAB)$$

Jaccard similarity measures the overlap between the sets of games carrying A and B. Hierarchical clustering is performed on Jaccard distance, $1 - J(A,B)$, using average linkage and a five-family cut. These are exploratory data-derived clusters, not official IGDB parent categories.

For country-level relatedness, the notebook first identifies the set S(c,p) of genres in which country c is specialized during period p under the RCA and minimum-count rule. For a pair of genres A and B, the period-specific proximity is the smaller of the two conditional probabilities of specialization across countries:

Equation (6). Symmetric genre proximity

$$\varphi(A,B,p) = \min\{P(B \text{ specialized} \mid A \text{ specialized}), P(A \text{ specialized} \mid B \text{ specialized})\}$$

Proximity is suppressed unless at least three countries specialize in each of the two genres in the period. The minimum makes the measure symmetric and prevents an asymmetric conditional probability from defining a high proximity by itself.

For a country c and a genre g in which it is not yet specialized, the relatedness score is the mean of φ(g,h,p) across the genres h in the country's current specialty set. The risk set contains country–genre observations for countries with game associations in both the current and next consecutive five-year periods, with at least one current specialty and with g not already specialized. The binary outcome equals one if the country is specialized in g in the next period. Adoption rates are summarized by quartile of the relatedness score, and a binomial generalized linear model is estimated with the relatedness score as the predictor and standard errors clustered by country. This is an observational association model; it does not control for all country, genre, or time-varying confounders.

4.5. Country trajectories and genre-entry sequences

Country portfolio breadth is the number of genres in which each country is specialized in each observed five-year period. Periods with no recorded game associations are retained as missing rather than coded as zero. A country must have observations in at least 60% of the study periods to receive a rule-based trajectory label. The notebook estimates a linear slope in the number of specialized genres per five-year period and distinguishes diversifying, narrowing/specializing, narrow-stable, broad-stable/mixed, and insufficient-coverage profiles according to predefined slope and level thresholds. K-means clustering on standardized trajectories is included as an exploratory analysis for countries with adequate period coverage; missing values are interpolated only for clustering and remain missing in the exported raw trajectory data.

For sequence analysis, each country's genre sequence is based on the first five-year period in which it becomes specialized in each genre. Genres first entered in the same period are treated as ties; no within-period order is imposed. The analysis counts transitions between successive distinct entry periods and summarizes each transition by the number of countries in which it occurs and its median lag in five-year periods. A separate "follower-like" timing profile records whether a country's first specialization in a genre occurs after the earliest country in the dataset. This is temporal ordering only and should not be interpreted as evidence of copying or influence.

4.6. Genre-combination link prediction

The link-prediction analysis represents the 23 genres as nodes in an undirected graph. An edge connects two genres if they occur together in at least one eligible game before the temporal cutoff. The candidate set consists of unordered genre pairs absent by 2015. For a candidate pair A,B, the primary Common Neighbors score is:

Equation (7). Common Neighbors score

$$CN(A,B) = |N(A) \cap N(B)|$$

The score counts the genres that have co-occurred with both A and B in the training graph. It is a ranking signal; it is not a probability that the pair will appear.

Three alternative heuristics are included as comparisons or tie-breakers: Jaccard similarity of the endpoint neighborhoods; Adamic–Adar, which sums the inverse logarithm of the degree of shared neighbors; and preferential attachment, which multiplies the degrees of the two endpoints. The 2015 model uses games from 1952–2015 for training and 2016–2025 for held-out evaluation. A candidate is positive if it appears in at least one held-out game. The primary ranking metric is hits@10, equivalently precision@10 when ten candidates are selected. A random baseline repeatedly draws ten candidates from the same candidate pool without replacement; 1,000 repetitions are used to estimate expected hits, the 95th percentile, and an empirical tail probability.

Equation (8). Precision at K

$$\text{Precision@K} = (\text{number of top-K candidates observed in test period}) / K$$

The empirical random-baseline p-value is the share of random draws with at least as many hits as the model, with a plus-one correction. With a small candidate pool, precision@10 and p-values can be unstable; the pool size and positive count must be reported.

To address saturation in the global pair graph, the analysis also ranks previously unseen three-genre combinations. A triple is positive if it appears in at least one held-out game. At the country level, a candidate is a genre pair not previously recorded for a country. The notebook evaluates countries with at least 20 pre-cutoff game records, constructs local country graphs, and ranks candidates using local common neighbors followed by global common neighbors and Adamic–Adar. The reported country-level random baseline draws within each country's own candidate pool. The final candidate lists use all observations through 2025 and are therefore unvalidated hypotheses for future observations. Rolling cutoffs at 2005, 2010, and 2015 are also generated for robustness, but their numerical results were not included in the summary supplied for this draft.

4.7. Software, reproducibility, and interactive data exploration

The data preparation and statistical analyses were implemented in Python notebooks executed through Google Colab, with the source data stored in a relational SQLite database. The computational workflow uses SQL and Python-based data processing to construct the analytical tables, calculate country–genre–year and country–genre–period measures, estimate the reported relatedness model, construct genre co-occurrence networks, and evaluate link-prediction rankings. The notebooks export intermediate tables, summary files, and figures so that the reported results can be checked against their underlying outputs. The notebooks and their parameter settings constitute the authoritative record of the analytical procedures reported in this paper.

To make the retained data easier to inspect, the authors also developed a browser-based filtering and visualization interface (available at https://game-research-analysis.vercel.app/). The interface provides a supplementary means of exploring the curated game records through the filtering and visualization functions implemented in the deployed application. It is intended to improve accessibility and support exploratory inspection of the dataset; it is not treated as an independent data source or as a replacement for the documented SQL and notebook workflow.

The distinction between the analytical pipeline and the interactive interface is important for reproducibility. All sample sizes, exclusion counts, derived variables, statistical estimates, and model-evaluation results reported in the manuscript should be generated from, or independently reconciled with, the archived database and notebook outputs. A view selected in the web interface may

help a reader examine a subset of records, but it should not be cited as the sole basis for a quantitative claim. Where the interface displays derived summaries, the corresponding calculation should be documented in the notebooks or in an accompanying technical specification.

For archival reproducibility, the authors should preserve a dated snapshot of the cleaned analytical data, the notebooks used for each research pillar, the configuration and filtering rules, and the exported tables and figures. The web application should be accompanied by a version identifier or release date and a short description of the filters and visualizations available in that version. Because a deployed web interface may change over time, the paper should record the access date and, where possible, provide a stable code repository or archived release alongside the live URL.

The interactive interface does not resolve the source database's missing or ambiguous metadata. In particular, it does not establish the semantic meaning of country associations, recover country information for games without a recorded company-country link, identify whether a platform association refers to an original release or a later port, or supply measures of sales, player counts, playtime, or country of play. These constraints remain part of the study's measurement limitations, regardless of how the data are presented or filtered in the interface.

The analysis also distinguishes implemented procedures from possible extensions. The reported notebooks provide descriptive cumulative-reach measures, revealed-specialization calculations, genre co-occurrence and relatedness measures, trajectory and sequence summaries, and temporally held-out link-prediction evaluations. Methods such as Bass or logistic diffusion-curve estimation, Cox survival analysis, NetInf, dynamic time warping, node2vec, matrix or tensor factorization, hidden Markov models, and gradient-boosted adoption prediction should be treated as future extensions unless they are separately implemented, evaluated, and reported.

5. Results

This section reports the descriptive summaries and model-evaluation statistics supplied from the three research notebooks. Results are presented in the same order as the research questions. Where the available summaries do not provide a numerical output, the text identifies the result as not yet available rather than inferring a value from the method. The counts in this section are metadata counts and should be interpreted within the country-field and sampling limitations described in Section 3.

5.1. Sample composition and genre representation

The cleaned dataset contains 75,971 eligible games associated with 118 countries and 23 genres over 1952–2025. The joined game–country–genre representation contains 158,939 rows. The genre vocabulary is relatively small, which makes exhaustive analysis of possible pairs and triples feasible: 253 unordered pairs and 1,771 unordered triples are possible among 23 labels.

The largest genre categories by distinct associated games are Adventure (24,461 games), Indie (18,517), Simulator (12,969), Role-playing (12,702), and Strategy (11,306). Their country counts differ from their game counts and refer to the number of country labels associated with at least one game in the relevant genre. These figures should not be read as the number of countries that crossed the stricter annual adoption threshold used in the diffusion analysis.

Table 5.1. Most represented genres by distinct associated games

Genre	Distinct games	Countries with ≥1 associated game
Adventure	24,461	103
Indie	18,517	109
Simulator	12,969	86
Role-playing (RPG)	12,702	88
Strategy	11,306	84

Note. The country column is an any-record presence count, not the threshold-based adoption reach reported in Table 5.2.

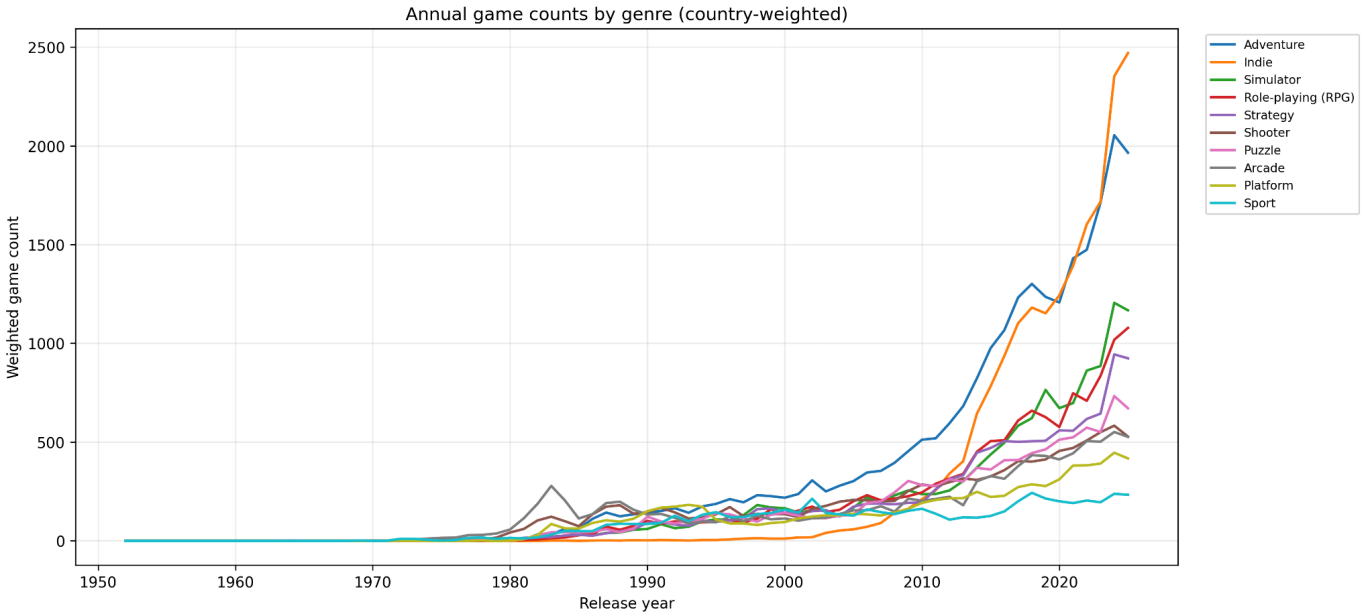
5.2. Temporal reach of genre categories

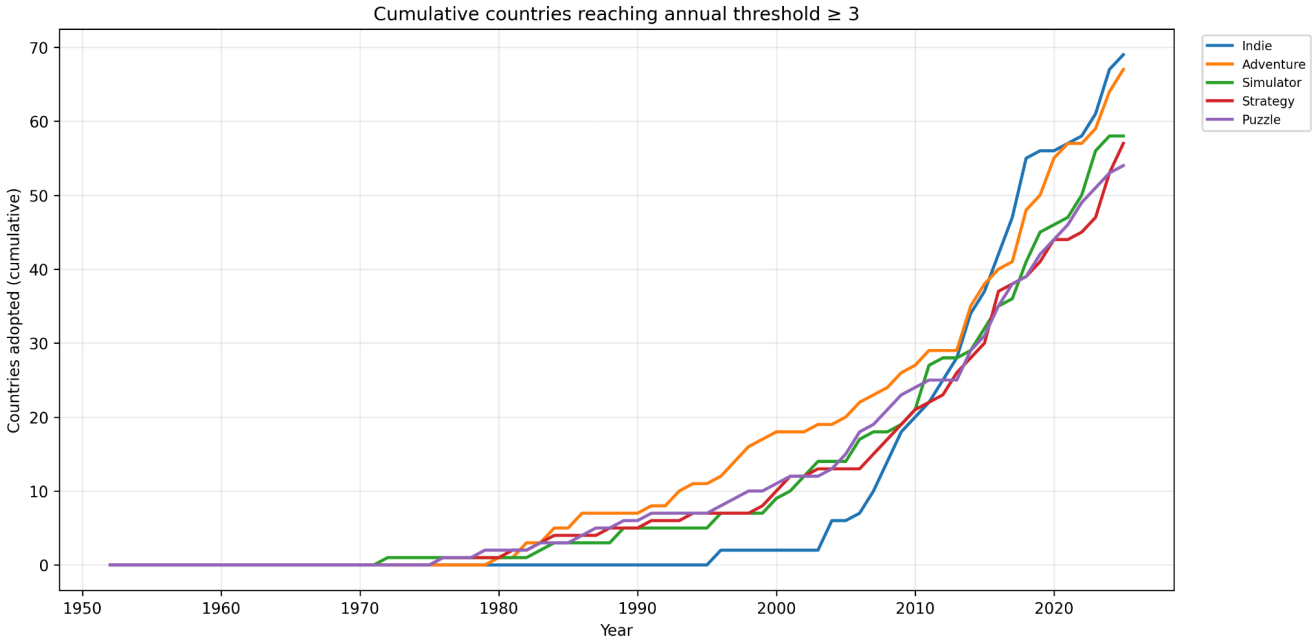
Under the annual weighted-count threshold of at least three, Indie reached the largest number of country-associated settings (69), followed by Adventure (67), Simulator (58), Strategy (57), and Puzzle (54). The earliest threshold-based appearances among these widely represented genres were recorded for Simulator in 1972, Strategy and Puzzle in 1976, Adventure in 1980, and Indie in 1996. The ordering shows that broad eventual reach and early appearance are distinct properties: Indie has the largest reach under the threshold, although its reported first threshold year is later than those of several other genres.

Table 5.2. Genres reaching the most countries under the annual threshold

Genre	Countries reaching threshold	First threshold year
Indie	69	1996
Adventure	67	1980
Simulator	58	1972
Strategy	57	1976
Puzzle	54	1976

Note. Operational appearance is the first year in which the country–genre weighted count reaches at least three. Test thresholds of two and five should be reported as robustness checks in the final manuscript.





The current summary does not provide the numerical peak years, fifth-country lags, decade-reach counts, or list of genres meeting the sustained-decline rule. Nor does it provide a table of fitted logistic/Bass curve parameters, because no such curve fit is implemented in the supplied diffusion notebook. Accordingly, no inference is made here about which categories follow statistically estimated S-curves or which have "died." These outputs should be added from the relevant CSV exports if they are to become part of the final results.

5.3. Country specialization and genre families

The country–genre specialization analysis uses 14 five-year periods anchored at 1952, with the final period partial and the latest complete period spanning 2017–2021. In that latest complete period, the

highest reported RCA values include Hungary–Pinball (117.17; 29 weighted credits), Czechia–Music (14.11; 103), China–MOBA (13.83; 6), Egypt–Point-and-click (13.75; 4), and Italy–Racing (7.49; 108). These examples illustrate why absolute counts must accompany relative indices. Hungary's very high Pinball RCA is based on 29 weighted credits, while some other high values occur with only three or four credits. A high RCA can reflect a small global or country denominator and should not be equated with high production volume.

Table 5.3. Selected high-RCA country–genre specializations, 2017–2021

Country	Genre	RCA	Weighted credits
Hungary	Pinball	117.17	29.0
Czechia	Music	14.11	103.0
China	MOBA	13.83	6.0
Egypt	Point-and-click	13.75	4.0
Armenia	Point-and-click	11.41	22.0
Italy	Racing	7.49	108.0
South Korea	MOBA	7.61	3.0

Note. Selected examples from the researcher-supplied top-RCA list. The specialty flag also requires RCA > 1 and at least 3 weighted game credits.

The co-occurrence analysis produced five exploratory genre families. One family contains 19 genres, while Music, Quiz/Trivia, Pinball, and MOBA each appear as a singleton family. The large cluster includes Adventure, Arcade, Card & Board Game, Fighting, Hack and slash/Beat 'em up, Indie, Platform, Point-and-click, Puzzle, Racing, Real Time Strategy, Role-playing, Shooter, Simulator, Sport, Strategy, Tactical, Turn-based strategy, and Visual Novel. This structure suggests that the selected Jaccard clustering does not separate most labels into sharply distinct groups; it may reflect broad co-occurrence among widely used labels and sparse or specialized co-occurrence for the singleton genres. It should not be presented as a definitive genre taxonomy.

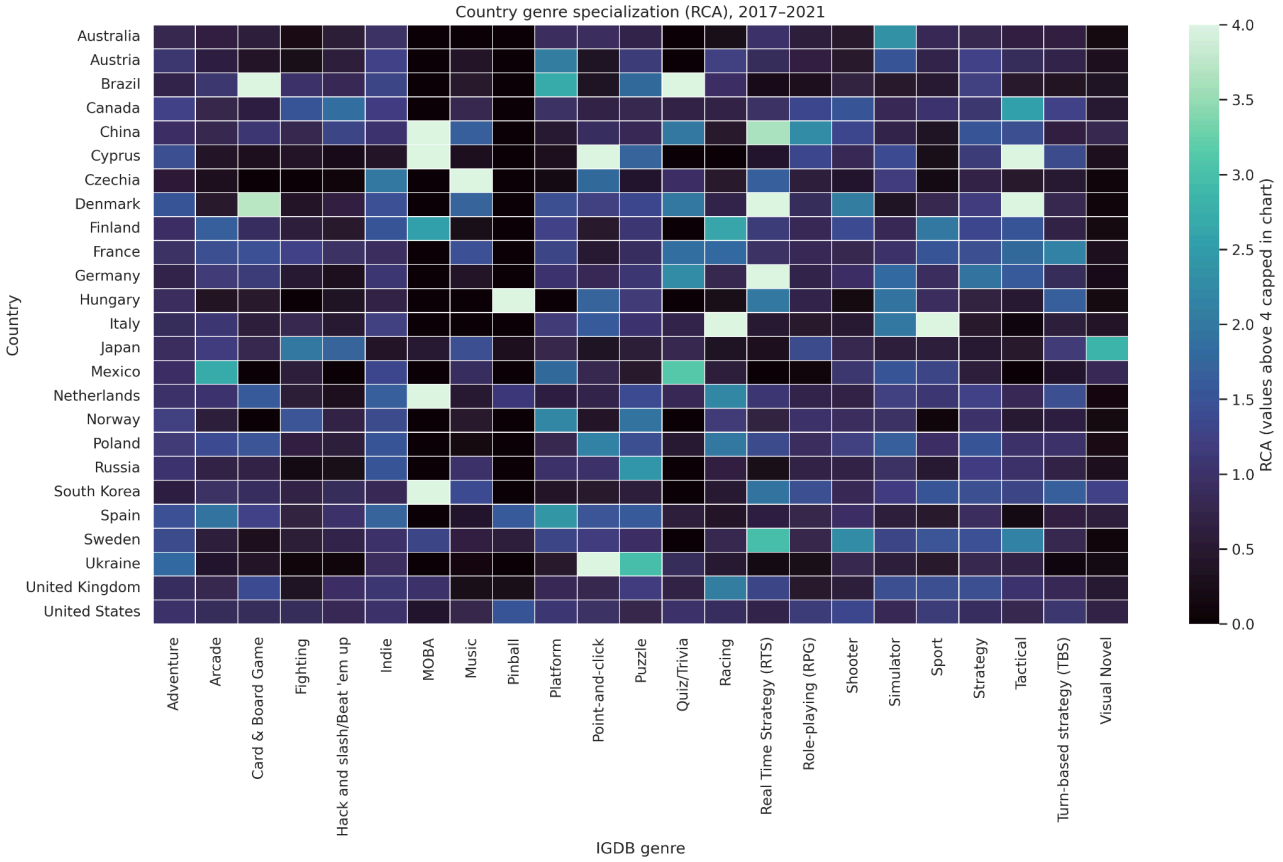
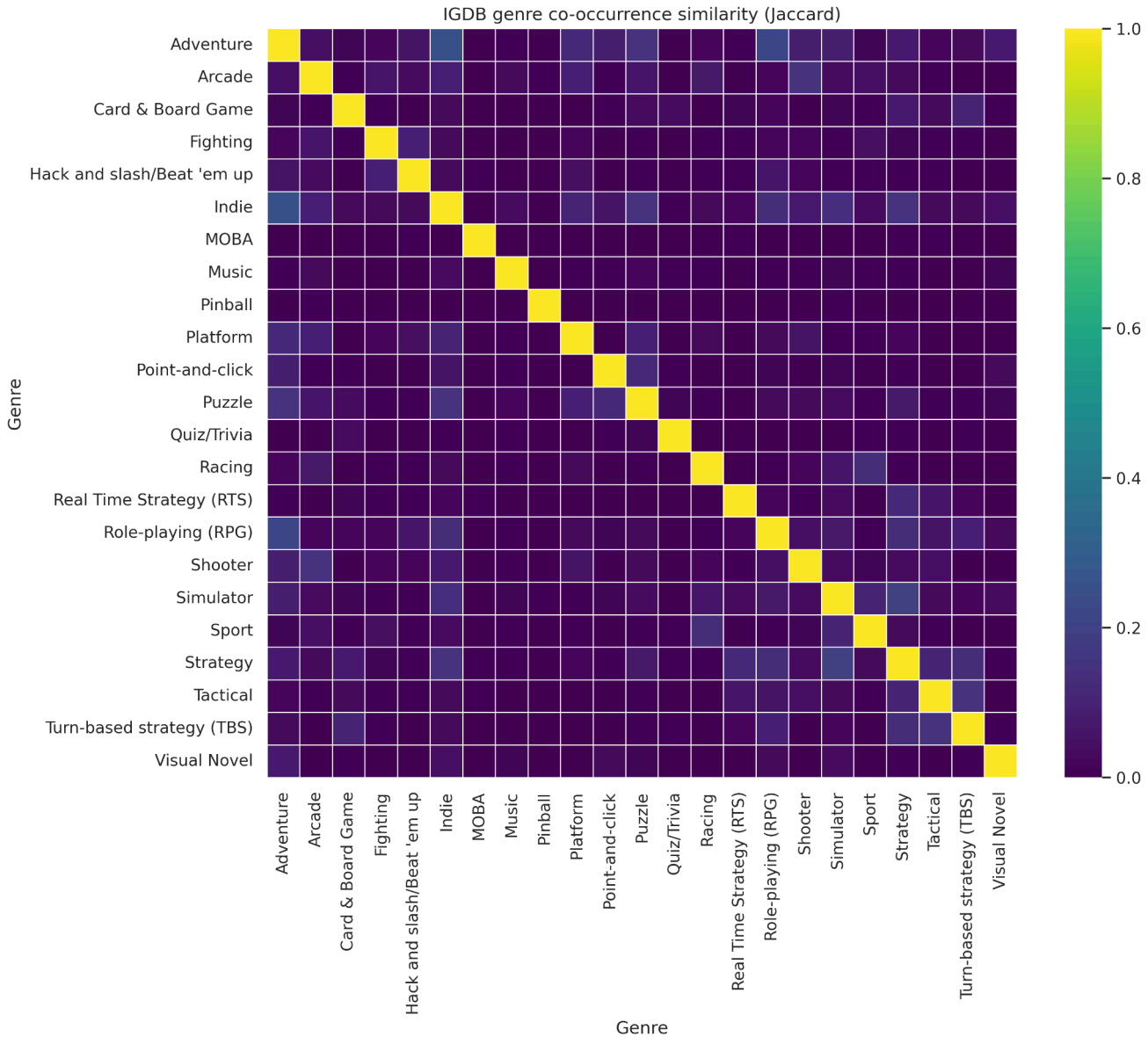


Table 5.4. Data-derived genre families from Jaccard hierarchical clustering

Cluster	Number of genres	Membership
Family 1	19	Adventure; Arcade; Card & Board Game; Fighting; Hack and slash/Beat 'em up; Indie; Platform; Point-and-click; Puzzle; Racing; Real Time Strategy; Role-playing; Shooter; Simulator; Sport; Strategy; Tactical; Turn-based strategy; Visual Novel
Family 2	1	Music
Family 3	1	Quiz/Trivia
Family 4	1	Pinball
Family 5	1	MOBA

Note. Five clusters were requested in the notebook. Clusters are based on average-linkage hierarchical clustering of Jaccard distances among game sets; they are not official IGDB families.



5.4. Relatedness and next-period specialization

The relationship between genre relatedness and subsequent specialization is monotonic across the four relatedness quartiles. In the lowest quartile, 7.4% of candidate country–genre observations became specializations in the next period (n = 1,047). The corresponding rates were 13.7% in Q2 (n = 1,047), 20.9% in Q3 (n = 1,046), and 28.6% in the highest quartile (Q4; n = 1,047). The difference between the highest and lowest quartiles is 21.2 percentage points. The pattern is consistent with the product-space expectation that new specializations are more likely when they are related to existing portfolio strengths.

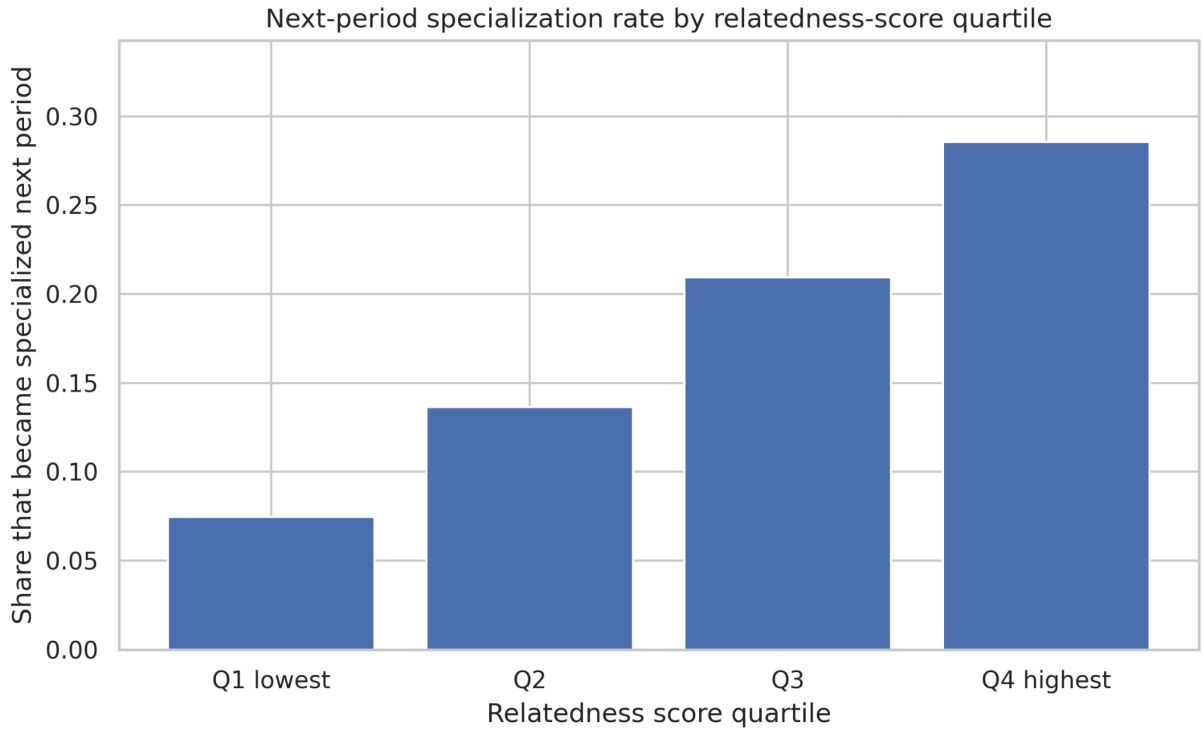
Table 5.5. Next-period specialization by relatedness quartile

Relatedness group	Candidate country–genre observations (n)	Next-period entry rate

Q1 — lowest	1,047	7.4%
Q2	1,047	13.7%
Q3	1,046	20.9%
Q4 — highest	1,047	28.6%

Note. Quartiles are based on ranked relatedness scores. The rates are descriptive transition proportions; country-period observations may be repeated within countries.

A binomial generalized linear model with country-clustered standard errors estimated an odds ratio of 119.409 per one-unit increase in the relatedness score (cluster-robust $p = 3.622 \times 10^{-25}$). The association is statistically strong in the reported model, but the magnitude of the odds ratio must be interpreted with attention to the scale and empirical distribution of the relatedness score. A full one-unit change may be much larger than a typical observed difference. For the paper, the quartile gradient is more directly interpretable; a rescaled coefficient (for example, per 0.1 standard deviation or interquartile-range change), its confidence interval, and model diagnostics should be added before emphasizing the odds ratio substantively.



5.5. Country trajectories and genre-entry sequences

The rule-based trajectory classification is limited by temporal coverage. Of the 118 represented countries, 102 are classified as having insufficient period coverage, 15 as diversifying, and one as broad/stable or mixed. The predominance of insufficient coverage means that the trajectory typology is informative only for a small subset of countries. It does not show that the remaining 102 countries are stable specialists; it shows that the data do not meet the notebook's minimum coverage requirement for assigning a trajectory class.

Table 5.6. Country trajectory classification

Trajectory category	Countries	Share of 118 represented countries
Insufficient period coverage	102	86.4%
Diversifying	15	12.7%
Broad / stable or mixed	1	0.8%

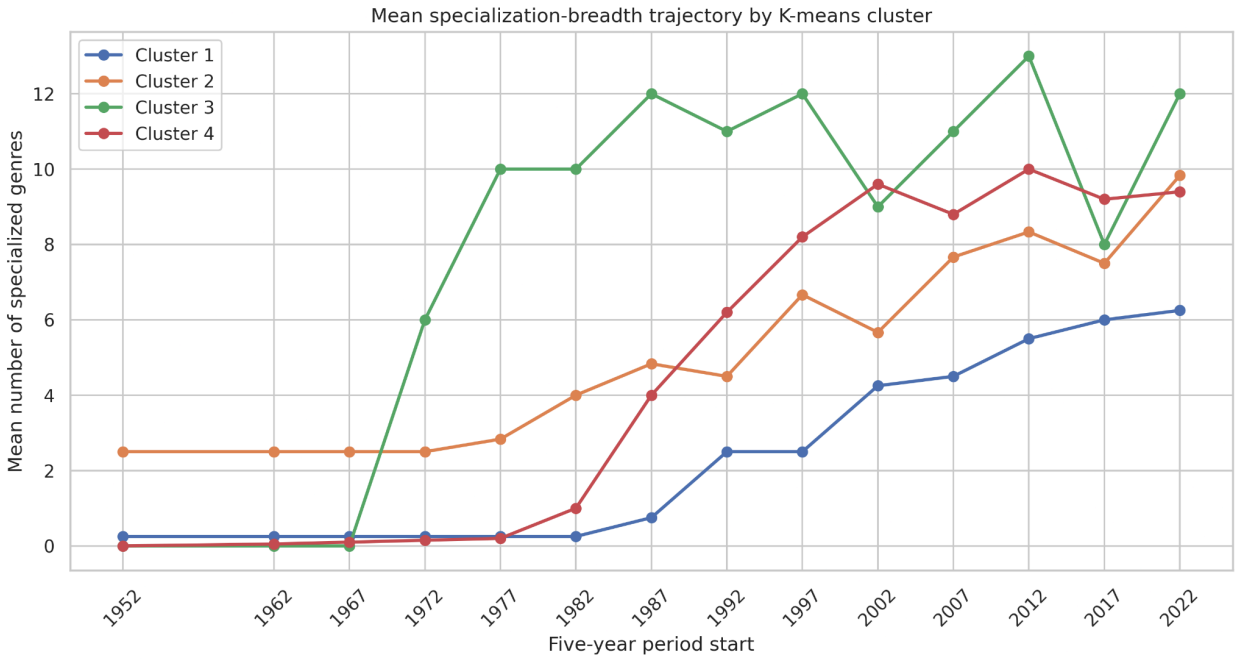
Note. Counts reproduce the automated summary. The notebook leaves periods without country-associated records missing and requires at least 60% period coverage for a substantive path label.

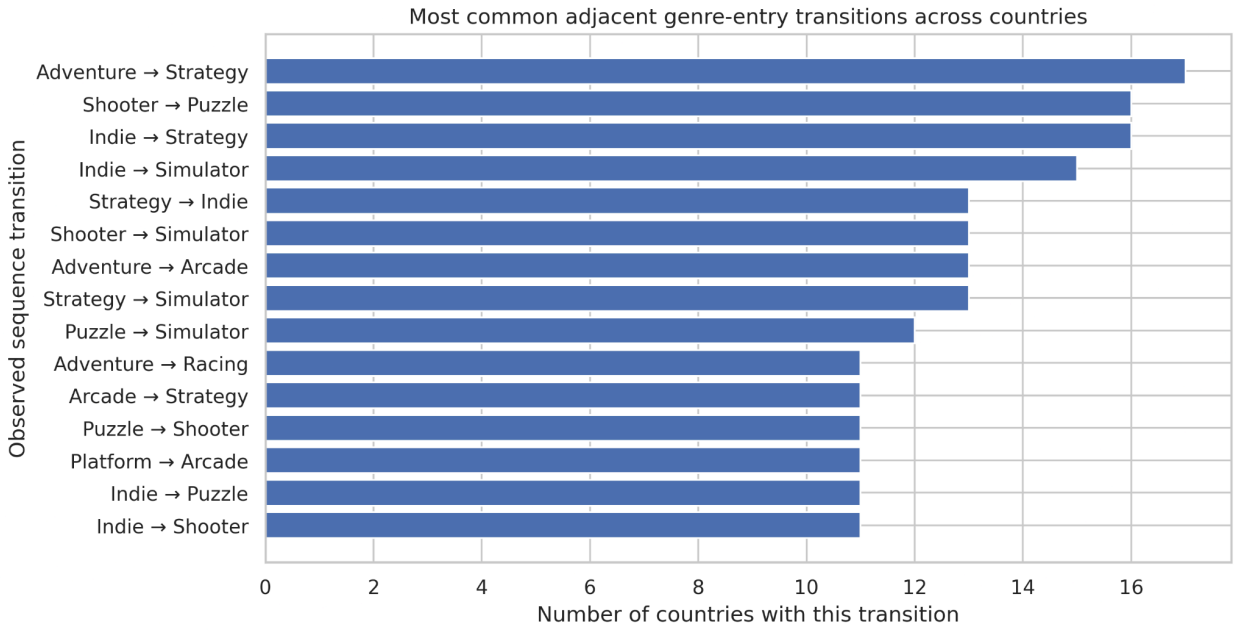
Among countries with sufficient evidence to define an entry sequence, the most common adjacent transitions are Adventure to Strategy (17 countries; median lag one five-year period), Indie to Strategy (16; one period), Shooter to Puzzle (16; one period), Indie to Simulator (15; one period), and Adventure to Arcade (13; one period). Other recurrent transitions include Shooter to Simulator, Strategy to Indie, Strategy to Simulator, Puzzle to Simulator, and Adventure to Racing. These sequences describe ordering among first specialization periods. Because five-year bins obscure within-period timing and simultaneous first entries are treated as ties, the results do not identify exact year-by-year transitions.

Table 5.7. Most frequent observed adjacent genre-specialization transitions

From genre	To genre	Countries	Median lag (5-year periods)
Adventure	Strategy	17	1
Indie	Strategy	16	1
Shooter	Puzzle	16	1
Indie	Simulator	15	1
Adventure	Arcade	13	1
Shooter	Simulator	13	1
Strategy	Indie	13	1
Strategy	Simulator	13	1
Puzzle	Simulator	12	1
Adventure	Racing	11	1

Note. A transition connects successive distinct first-specialization periods within a country. One period corresponds to five years; ties within the same period do not generate an ordered transition.





5.6. Prediction of new genre combinations

The 2015 cutoff experiment highlights a limitation of global pair prediction. By the end of 2015, 240 of the 253 possible genre pairs had already appeared, leaving only 13 candidate pairs that were genuinely unseen in the training period. Six of these 13 candidates appeared at least once between 2016 and 2025. The Common Neighbors model placed five of the six future positives in its top ten, yielding precision@10 of 0.50. However, random selection from the same candidate pool produced an expected 4.598 hits, with a 95th-percentile value of six hits. The empirical random-baseline p-value was 0.5594.

Thus the top-ranked global pair list did not outperform random selection convincingly under this evaluation design.

Table 5.8. Global link prediction for previously unseen genre pairs, 2015 cutoff

Metric	Result
Possible genre pairs	253
Pairs observed by 2015	240
Missing pair candidates	13
Candidates appearing in 2016–2025	6
Common Neighbors hits@10	5
Precision@10	0.500
Random expected hits	4.598
Random 95th percentile	6
Empirical random-baseline p-value	0.5594

Note. A positive candidate is a pair absent by 2015 that appears together in at least one game during 2016–2025. The small candidate pool limits the precision of the test.

The triple experiment provides a larger candidate space. Of 1,771 possible unordered triples, 1,081 were observed by the cutoff, leaving 690 unseen candidates. A total of 226 of these unseen triples appeared during 2016–2025. The top-ten triple ranking returned four observed combinations, corresponding to precision@10 of 0.40. No random-baseline statistic for the triple task was included in the supplied summary, so the result cannot be described as statistically superior to chance.

Table 5.9. Triple-combination prediction, 2015 cutoff

Metric	Result
Possible unordered triples	1,771
Triples observed by 2015	1,081
Missing triple candidates	690
Candidates appearing in 2016–2025	226
Top-ten hits	4
Precision@10	0.400

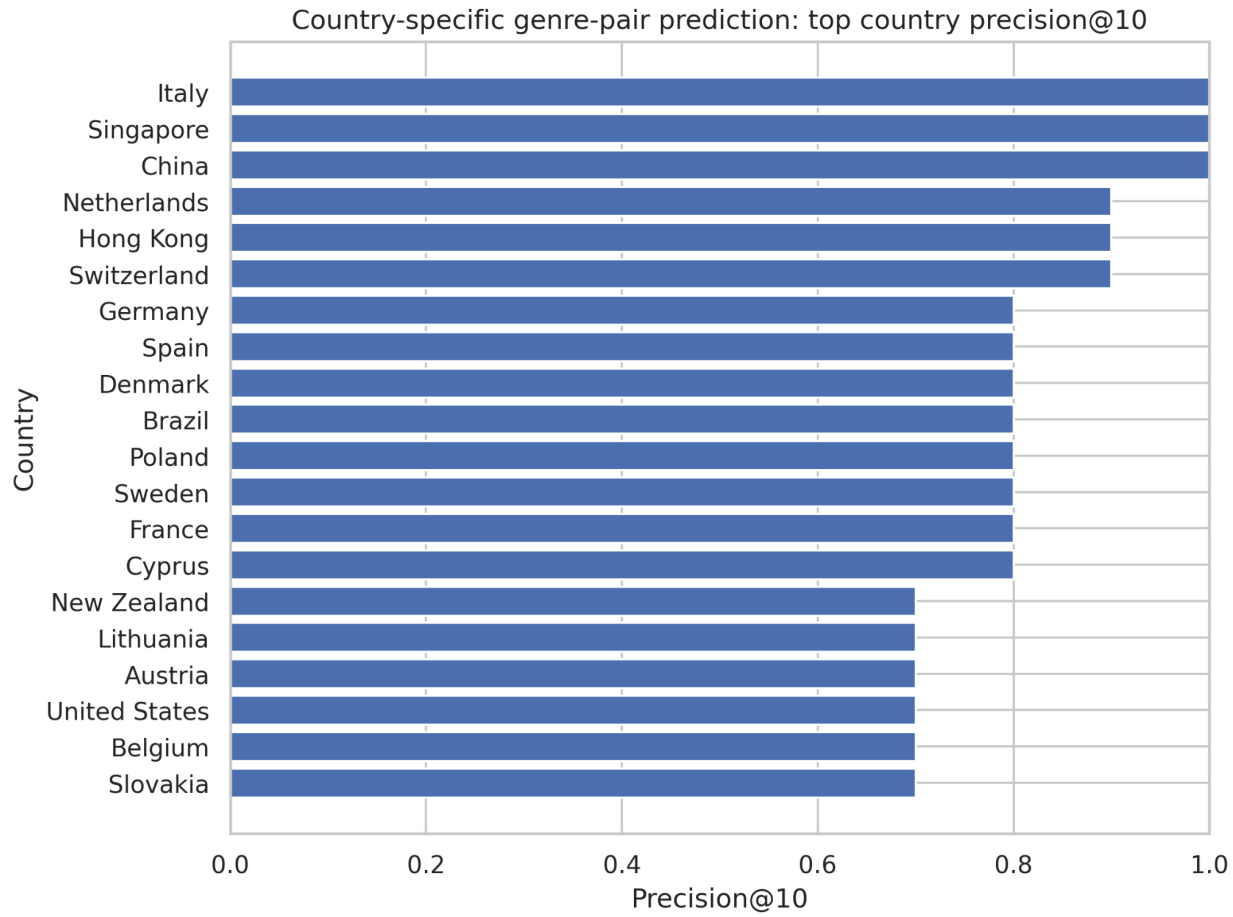
Note. A game with four genre labels contributes four distinct three-genre subsets. Triple events are therefore overlapping observations.

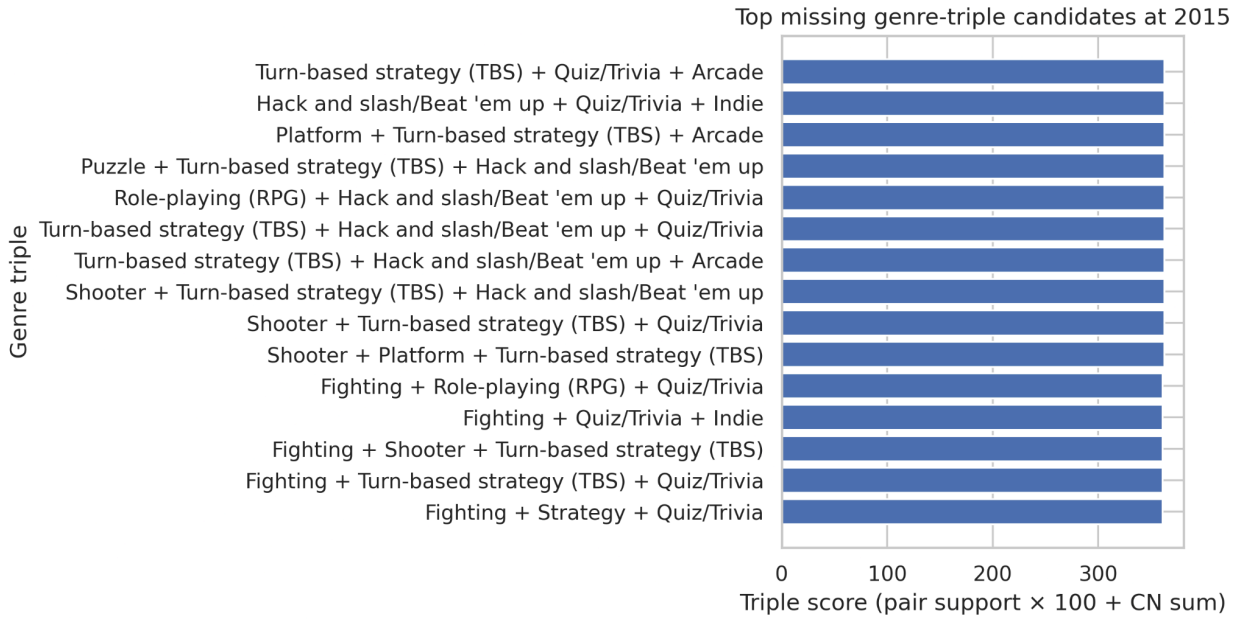
The country-specific pair experiment produced a stronger preliminary result. The evaluation covered 47 countries and 8,641 country–pair candidates. Across 47 sets of ten ranked predictions, 297 of 470 selected slots were hits, for an overall precision@10 of 0.6319. The random baseline averaged 122.335 hits, and its 95th percentile was 137.05; the empirical p-value was 0.000999. The result indicates that local and global co-occurrence structure ranks country-specific combinations more effectively than random selection in this test. It should nevertheless be interpreted as preliminary because it depends on country-specific candidate construction, a minimum of 20 pre-cutoff games per country, repeated evaluation slots across countries, and the unresolved semantics and completeness of country associations.

Table 5.10. Country-specific genre-pair prediction, 2015 cutoff

Metric	Result
Countries evaluated	47
Country-specific candidates	8,641
Top-K slots (10 per country)	470
Observed hits	297
Precision@10, pooled	0.6319
Random mean hits	122.335
Random 95th percentile	137.05
Empirical random-baseline p-value	0.000999

Note. Countries were included when they had at least 20 pre-cutoff game records. The candidate ranking uses country-local common neighbors, then global common neighbors and Adamic–Adar. The random baseline samples from each country's own candidate set.





Finally, the complete 1952–2025 graph leaves 13 global pairs and 690 global triples unobserved in the retained genre vocabulary. Their ranks are candidate hypotheses for tracking after 2025, not validated forecasts. Because most pairs already exist, the more informative predictive contribution may lie in higher-order combinations and country-conditional novelty rather than in the saturated global pair graph.

6. Discussion

6.1. Three complementary views of category evolution

The findings suggest that genre evolution is better represented as a combination of temporal reach, portfolio structure, and recombination than as a single diffusion curve. The diffusion results show that high eventual reach and early threshold crossing do not coincide perfectly: Indie has the largest reported threshold-based reach, while Simulator appears earliest among the five leading categories. The specialization results show that countries differ in their relative genre portfolios and that transitions into new specialties are more frequent when the candidate genre is related to the country's existing specialties. The link-prediction results, in turn, reveal that global pair novelty is limited by the near-saturation of the 23-genre vocabulary, while country-specific combinations remain more open to prediction.

Together, these patterns support an interpretation of genre labels as a relational system. Categories are not simply adopted independently; they coexist within games, form overlapping genre neighborhoods, and appear within country-associated portfolios that vary through time. This relational framing connects genre analysis to work on multidimensional classification and economic complexity (Faisal & Peltoniemi, 2018; Hidalgo et al., 2007), while remaining grounded in a specific metadata corpus.

6.2. Diffusion as observed reach rather than demonstrated transmission

The cross-country reach measures provide a systematic way to compare how widely a genre is represented across country labels. However, the interpretation depends critically on the country field. Until it is confirmed whether the country association denotes developer origin, publisher location, release

territory, or another relation, the results should be described as the spread of country-associated records, not the spread of national production or consumer adoption. The threshold of three weighted records is a transparent operational rule, but it is not a natural boundary between adoption and non-adoption. The sensitivity analysis at thresholds two and five should be included in the final article to show whether genre rankings are robust to this choice.

The original proposal was interested in "invisible pathways" linking countries through geography, language, corporate networks, and platforms. The present results do not identify those pathways. They establish temporal precedence and differences in country reach, which can be useful descriptive foundations, but they cannot show that one country transmitted a design idea to another. This distinction follows diffusion theory's emphasis on channels and social systems: the current data observe outcomes that may be consistent with diffusion but do not observe the communication process itself (Rogers, 2003; Granovetter, 1973).

6.3. Relatedness and the path dependence of country portfolios

The increase in next-period specialization rates from 7.4% in the lowest relatedness quartile to 28.6% in the highest is the clearest result linking the country-portfolio analysis to the economic-complexity literature. It is consistent with a path-dependent process in which existing specializations are associated with a greater likelihood of entering related categories. This parallels the product-space finding that countries tend to diversify toward products connected to their current export basket (Hidalgo et al., 2007; Hausmann & Klinger, 2007).

The analogy should not be overstated. The country-genre RCA in this study is calculated from metadata associations, not exports or measured productive capabilities. A country may have a high RCA because a genre occupies a relatively large share of a small number of recorded associations. The minimum weighted-count rule mitigates but does not eliminate this denominator sensitivity. A defensible interpretation is therefore that country-associated genre portfolios exhibit a relatedness gradient in subsequent relative specialization. Whether this reflects skills, supply chains, institutional capabilities, market demand, tagging conventions, or database coverage remains unresolved.

The trajectory results also temper the broader diversification narrative. Only 16 of 118 represented countries received a substantive trajectory label, while 102 failed the minimum temporal-coverage rule. The available classification therefore does not support a general conclusion that countries are diversifiers, specialists, or followers. A more complete account requires either better historical coverage or a narrower sample of countries observed consistently across the period.

6.4. Genre families and the limits of co-occurrence taxonomies

The five-cluster solution—one 19-genre family and four singletons—does not produce a finely differentiated taxonomy. Rather than forcing a substantive interpretation onto this structure, it should be treated as evidence that the chosen clustering procedure and Jaccard distance yield one broad co-occurrence core and a small set of labels with comparatively distinct game sets. This may reflect true structural separation, low frequency, limited overlap, or the imposed number of clusters. It also reinforces the conceptual point that a data-derived family is not equivalent to an official taxonomy or a stable genre ontology (Apperley, 2006; Starosta et al., 2024).

A useful next step is to report the full similarity matrix and dendrogram, compare more than one cluster cut, and test whether the singleton assignments persist under minimum co-occurrence thresholds and alternative similarity measures. PMI should be reported with support counts because rare co-occurrences can receive large association scores. The family results are therefore exploratory and best used to motivate taxonomic comparison, not to assert that the genre system contains five natural families.

6.5. Why country-specific link prediction is more promising than global pair prediction

The global pair experiment is constrained by a small candidate set: 240 of 253 possible genre pairs had already appeared by 2015. In this setting, the Common Neighbors ranking produced five hits in the top ten, but random selection produced nearly as many on average and the empirical p-value was not significant. This is not evidence that network heuristics are generally ineffective; it is evidence that this particular global prediction task has limited headroom because the genre vocabulary is small and the graph is already dense.

The country-specific task changes the prediction space. A pair that exists somewhere in the global graph may still be new within a country's historical associated portfolio. This creates a larger set of possible country–pair links and allows country-local neighborhoods to provide predictive information. The observed precision of 0.632 compared with a random expectation of approximately 122 hits across 470 slots suggests that the ranking contains useful signal in the supplied evaluation. This result is promising but requires stronger validation before it is treated as a general predictive finding. In particular, country-level outcomes should be tested using rolling temporal cutoffs, minimum support thresholds, alternative country inclusion rules, and metrics that account for the number of positive candidates available in each country.

The triple experiment provides an additional response to global pair saturation. Its top-ten precision of 0.40 is a descriptive result, but without a reported triple-specific random baseline it is not possible to state whether it is better than random. Future versions should benchmark triples under the same candidate-pool randomization procedure and report recall, average precision, and calibration if the output is to be interpreted probabilistically. More generally, link-prediction scores should be understood as rankings of plausible co-occurrences, not as estimated probabilities of innovation.

6.6. Contribution, limits, and next steps

The study contributes a reproducible longitudinal framework for examining a multi-label category system through three linked lenses: country-associated diffusion and reach, relative country–genre specialization and relatedness, and temporal prediction of new genre combinations. Its broader value lies in the methodological transfer of tools from economic complexity and network science to cultural-product metadata. It also demonstrates why negative or inconclusive findings matter: global pair prediction is not convincingly better than random under the current candidate set, and the country trajectory analysis is constrained by extensive missing temporal coverage.

The most important limitation is the unresolved semantics of the country association. This must be documented before the manuscript uses terms such as developer country, national production, or adoption. Additional limitations include uneven temporal coverage, selection into the IGDB corpus, potentially inconsistent genre tagging, multi-label dependence among pair and triple events, small numbers of unseen global pairs, and the preliminary nature of the country-specific prediction test. The current analysis is

observational and does not estimate causal effects of geography, language, firms, publishers, platforms, or cultural proximity.

The next stage should prioritize data validation over methodological complexity. First, verify the country field and reconcile the historical sample-selection counts. Second, export and inspect the full diffusion summary (including peak years, fifth-country lags, decline flags, threshold sensitivity, and lead–lag network results). Third, add confidence intervals, rescaled relatedness coefficients, and diagnostic checks to the GLM. Fourth, evaluate link prediction at multiple temporal cutoffs and with alternative support thresholds. If the research later acquires studio, publisher, platform, language, geographic-distance, and country-industry variables, it can test the structural, technological, and cultural mechanisms proposed in the original concept note. Until then, the paper's claims should remain focused on the temporal and relational regularities observed in the IGDB-derived metadata.

In this sense, the realized study does not abandon the initial proposal; it refines it to match the evidence currently available. It moves from a broad causal account of how innovations cross borders toward a measurable account of how categories appear, become relatively specialized, and recombine across a long historical record. That narrower claim is more defensible and provides a foundation for subsequent mechanism-based research.

7. Limitations and Future Research

The findings should be interpreted in light of substantial constraints in the coverage, semantics, and temporal comparability of the underlying metadata. These constraints do not make the analysis uninformative; rather, they define the population to which its conclusions can reasonably apply. The study describes patterns among games with usable country and genre associations in the retained IGDB-derived dataset. It does not provide a complete census of game development, national production, consumer demand, or cultural influence.

7.1. Selection into the analytical sample

The initial collection contained more than 269,000 game records, whereas the analytical dataset contains 75,971 games. Retention required a record to have both a usable country association and at least one genre association. A substantial number of initially collected titles could not be retained because one or both forms of metadata were absent. In the researcher's inspection, many records appeared to involve individually developed titles or otherwise lacked a country association that could be used in the analysis. The data do not, however, establish that every country-missing title was independently developed; missing country metadata should not be treated as a reliable indicator of studio size or organizational form.

This filtering creates a potentially non-random selection problem. Titles with better-documented studios, publishers, release histories, or database records may be more likely to enter the analytical sample. Less visible independent projects, small-market releases, older titles, and games with incomplete metadata may be underrepresented. The resulting genre and country distributions therefore describe the retained metadata population rather than all games released between 1952 and 2025. Because missingness may be related to the scale, age, origin, or documentation of a title, the direction and magnitude of this selection bias cannot be inferred from the retained sample alone.

A final manuscript should include a transparent sample-flow diagram showing the number of records at each filter and the number excluded for missing country, missing genre, or both. The earlier proposal-level counts and the current retained sample should be reconciled from the original extraction logs before publication. Where possible, the analysis should compare retained and excluded records on variables that remain available, such as release year or platform, and report annual retention rates. Such comparisons would help determine whether historical periods or particular classes of games are disproportionately lost during filtering.

7.2. Country attribution and the meaning of geographic "adoption"

The most consequential construct-validity limitation is the unresolved meaning of the country association in the source data. A country label may refer to a developer, publisher, company headquarters, production relationship, licensing arrangement, release territory, or another database-defined association. These meanings are not interchangeable. Until the exact field provenance and relationship semantics are confirmed, the paper should use neutral expressions such as "country-associated game records," "country-level genre representation," and "threshold-based first appearance." It should not describe the country field as developer origin, national production, consumer adoption, or market demand without supporting documentation.

This ambiguity also limits the interpretation of the diffusion analysis. The first year in which a country reaches three weighted records is an operational first appearance in the retained metadata. It does not necessarily identify where a genre was invented, where it first became popular, or where players first encountered it. A hybrid genre may have been designed, introduced, or popularized by an individual developer whose country is absent from the database. If that early title is missing or geographically unassigned, the observed first appearance may occur later and in a different country. Consequently, the data cannot reconstruct the complete origin-to-destination pathway of a genre innovation. They can compare the timing and reach of the country associations that are recorded.

The fractional weighting rule reduces the inflation that would arise when a game is associated with multiple countries: a game linked to k countries contributes 1/k to each. This is a transparent accounting convention, but it assumes equal attribution across the listed countries. It cannot represent unequal contributions by co-developers, outsourcing partners, publishers, or subsidiaries. Sensitivity analyses using unweighted country associations and alternative attribution rules would show how much the results depend on this assumption.

7.3. Genre-label incompleteness and classification dependence

The same filtering issue applies to genre metadata. Games without an attached genre label could not contribute to the genre-level analyses, even when the title itself may have contained recognizable mechanics or genre elements. The retained 23-label vocabulary is therefore a source-defined and coverage-dependent representation of the field, not an exhaustive ontology of game design. Database labels may be broad, overlapping, inconsistently assigned, or added retrospectively. A hybrid game may be represented by only one label, while another may receive several. This affects genre counts, co-occurrence networks, country specialization, and the apparent emergence of new combinations.

The observed co-occurrence structure reinforces this caution. The exploratory family analysis placed 19 genres in one large family and left Music, Quiz/Trivia, Pinball, and MOBA as singletons. This pattern

may reflect the broadness of the available labels, density of common combinations, low support for rarer genres, or the selected similarity and clustering thresholds. It should not be interpreted as evidence that the 19 labels form a single substantive genre family. Future work should compare alternative co-occurrence measures, thresholds, and clustering procedures, and, where available, validate the results against finer-grained tags or expert-coded descriptions.

Multi-label coding also means that genre shares are not mutually exclusive. A title assigned to three genres contributes to all three genre counts; consequently, genre-specific totals may exceed the number of unique games. Pair and triple events are overlapping subsets rather than independent observations. Inference and uncertainty estimates should account for this dependence where the chosen model permits it.

7.4. Historical platform change and temporal comparability

Platform metadata presents a further historical measurement problem. A game may have been designed and first released for an older device, then ported, remastered, reissued, or made available on a newer platform years later. If the database records all associated platforms without distinguishing the original release platform from later platform availability, treating platform association as a contemporaneous development variable can assign an old title to a hardware generation that did not exist when the game was created. A simple comparison of platform prevalence across years would then conflate original technological context with later distribution and re-release histories.

For this reason, the present paper does not use platform associations to explain genre diffusion or national specialization. A stronger platform analysis would need, at minimum, a validated first-release platform and date, separate records for ports and re-releases, and an explicit definition of platform generation. It would also need to consider changes in the commercial and technological roles of consoles, computers, handhelds, mobile devices, and online distribution. These data could support a future analysis of whether platform transitions are associated with changes in genre diversity or the diffusion of combinations, but they cannot be inferred safely from an undifferentiated list of platforms.

7.5. Metadata appearances are not measures of use or market adoption

The dataset does not provide a reliable, comprehensive measure of how much each game was played, by whom, or in which country. Release records and genre tags therefore cannot establish audience exposure, player uptake, playtime, downloads, sales, revenue, active users, or cultural popularity. A title may be listed in a country association but have few players there; conversely, a game may be widely played in a country even when no corresponding country association is recorded. The adoption threshold used in Idea 1 should consequently be interpreted as a rule for identifying repeated metadata representation, not as a threshold of consumer adoption.

This distinction is especially important because use-based adoption may differ substantially from release-based or catalogue-based appearance. The lack of comprehensive game-level playtime and country-of-play data prevents the study from testing whether a genre became popular among consumers or whether usage spread from one national market to another. Future research could integrate compatible sales, player, platform, or playtime data, provided that country attribution and historical coverage are sufficiently reliable and that the different sources can be linked without introducing severe measurement error.

7.6. Temporal coverage, thresholds, and model uncertainty

The 1952–2025 observation window is long, but the completeness of records is unlikely to be uniform across decades. Older games may be less comprehensively catalogued, and the probability that a title has detailed country, genre, and platform information may vary by era. Apparent growth in genre counts can therefore reflect both changes in the underlying industry and changes in the coverage of the database. Annual counts, first appearances, peak years, and decline rules should be interpreted alongside year-specific sample coverage. The five-year periods used in the specialization analysis also smooth short-term changes and can place several entries in the same period, making their within-period order unknowable.

Several results depend on operational thresholds. The annual first-appearance rule uses at least three weighted country-game records; the specialization rule uses RCA greater than one and a weighted count of at least three; the decline definition uses a period below 25% of peak activity for five consecutive years. These rules improve reproducibility but are not natural boundaries. Threshold sensitivity should be reported, especially for low-volume countries and genres. Very high RCA values can arise when a genre has a small country-period denominator, so RCA should always be shown with its underlying weighted count and interpreted as relative representation within this sample rather than as a comprehensive measure of economic comparative advantage.

The country-trajectory classification is particularly constrained by missing temporal coverage: 102 of 118 countries were classified as having insufficient period coverage, while 15 were classified as diversifying and one as broad/stable or mixed. The 102 countries should not be interpreted as non-diversifiers or persistent specialists. The observed classifications apply only to the subset with sufficient usable period data. Similarly, the relatedness gradient is an observational association. The reported odds ratio of 119.409 is expressed per a full one-unit increase in the relatedness score and may be difficult to interpret without knowing the score distribution and rescaling. The quartile adoption rates provide a more intuitive descriptive comparison, but neither the gradient nor the GLM establishes that relatedness causes entry.

7.7. Limits of link-prediction evaluation and causal inference

The global pair-prediction task is constrained by a saturated candidate universe: 240 of 253 possible pairs had already appeared by the end of 2015, leaving only 13 unseen global candidates. With so few candidates, Hits@10 and Precision@10 are inherently unstable and provide little room for a sophisticated ranking to outperform a random draw. In the supplied evaluation, Common Neighbors returned five hits in the top ten, but the random baseline expected 4.598 and the empirical p-value was 0.5594. This is not evidence of convincing global predictive superiority. The triple task has a larger candidate space, but the reported top-ten precision of 0.40 should not be described as better than chance without a triple-specific random baseline.

The country-specific pair result is more promising, but it remains preliminary. It covers 47 countries and 8,641 country-pair candidates, with 297 hits among 470 top-ranked slots and an empirical random-baseline p-value of 0.000999. Its interpretation depends on the country sample inclusion rule, candidate construction, country-level support, repeated ranking slots, and the semantics of the country association. Rolling temporal cutoffs, per-country performance distributions, alternative minimum-support thresholds, and evaluation metrics that account for class imbalance are needed before generalizing this result.

Finally, the current analysis does not test the mechanisms proposed in the initial research proposal—geographic distance, shared language, cultural proximity, publisher or studio networks, or platform pathways—because the analytical dataset does not contain validated measures of these factors. Nor does it identify causal effects. Temporal precedence, relatedness, and predictive ranking are compatible with several explanations, including metadata practices and selection into the database. Future studies should add external firm, platform, language, geography, market, and user-behaviour data, then use designs appropriate to the causal question rather than treating observed sequence as evidence of transmission.

7.8. Priorities for future research

The most valuable next steps are data-oriented. First, document the exact meaning of each country relationship and determine whether multiple country associations refer to co-development, company location, publisher location, or another construct. Second, preserve records without country or genre associations in an exclusion file so that missingness can be described rather than silently discarded. Third, distinguish first-release platform from later ports and re-releases, and seek reliable country-level usage, sales, or playtime observations if the research question is extended from metadata appearance to consumer adoption. Fourth, improve temporal validation by testing multiple historical cutoffs and reporting uncertainty and country-level dispersion. Finally, where a suitable taxonomy or richer tag source is available, repeat the co-occurrence and link-prediction analyses with finer categories and compare the resulting structures.

These extensions would reconnect the realized empirical design to the original proposal's interest in structural, technological, and cultural pathways. They would also permit stronger statements about where genre innovations originate, which channels carry them, and whether the appearance of a genre in a country corresponds to actual player uptake. Until those data are available, the present study is best understood as a historical analysis of the recorded representation and recombination of genre categories across country associations.

8. Conclusion

This study examines the historical evolution of video-game genre categories through three complementary empirical perspectives: the temporal and geographic reach of genres, the changing specialization profiles of country-associated game records, and the emergence of new genre combinations. Using an IGDB-derived dataset of 75,971 eligible games released between 1952 and 2025, representing 118 countries and 23 genre labels, the analysis treats genres as overlapping categories whose representation can be traced across time and whose relationships can be studied through co-occurrence networks.

The diffusion analysis shows that genre reach and timing vary across categories. Under the operational threshold of three weighted records, Indie appears in the largest number of represented countries (69), followed by Adventure (67), while Simulator has the earliest first recorded country appearance among the leading genres (1972). These findings describe the temporal distribution of country-associated metadata; they do not establish the geographic origin of a genre or the movement of consumer demand. The country-specialization analysis identifies relative portfolio differences and a clear descriptive relatedness gradient: next-period specialization rates rise from 7.4% in the lowest relatedness quartile to 28.6% in the

highest. This pattern is consistent with path-dependent portfolio development, but it remains observational and is not equivalent to evidence about national productive capabilities or causal learning.

The link-prediction analysis produces a differentiated result. Global pair prediction has limited headroom because nearly all possible pairs in the 23-genre vocabulary had already appeared by 2015; its top-ten ranking does not outperform the random baseline convincingly. Country-specific pair prediction is more promising in the supplied holdout evaluation, although it requires additional temporal and country-level robustness checks. The triple analysis provides a larger space of candidate combinations but needs a matched random benchmark before its ranking performance can be evaluated inferentially.

The paper's principal contribution is therefore methodological and empirical. It brings together diffusion measures, revealed specialization, relatedness, sequence analysis, and network link prediction to study a long-run multi-label category system. It demonstrates how relational metadata can be used to map patterns of category prevalence, portfolio structure, and recombination, while also showing why selection, missing attribution, and taxonomy design must be treated as central measurement issues rather than peripheral data-cleaning concerns.

The scope of the conclusions is deliberately bounded. The current country field is not yet verified as a measure of developer origin or consumer market; the retained sample excludes many records without country or genre associations; platform histories may conflate original releases with later ports; and comprehensive country-level playtime or usage data are unavailable. These constraints prevent a complete reconstruction of where genre innovations began, how they travelled through firms and platforms, or when audiences adopted them. The next stage is consequently not simply to add more complex models, but to improve the underlying attribution and usage data. With those improvements, the framework developed here can support a more direct investigation of the structural, technological, and cultural channels through which creative innovations are developed, recombined, and circulated across national settings.

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